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Research Article

Bubble Size Distribution in Oil-Based Bubble Columns

A practical population balance model was used to evaluate the bubble size distribution in a bubble column. In addition, the bubble size distribution in the bubble column was measured at different gas velocities by photography and analysis of the pictures. Four types of liquid, i.e., water and three petroleum-based liquids, were used in the experiments. The gas phase was air. It was found that the existing models in the literature are not able to satisfactorily predict the experimentally measured bubble size distribution. The model can be corrected by applying a correction factor to the energy dissipation rate. The corrected model fits the experimental bubble size distribution considerably better than the existing models. The variation of this correction factor is reported for different systems at different gas velocities.

Keywords: Bubble columns, Coalescence, Modeling

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1 Introduction

Bubble size distribution is one of the most basic characteristics of bubble column reactors. Bubble properties, e.g., size and concentration, are decisive factors affecting the hydrodynamics and transport properties in the column. From obtaining knowledge of the size distribution of the bubbles, averaged properties such as interfacial area or Sauter mean diameter values, which are often required in engineering problems, can be determined [1]. The most important deterministic model for describing bubble size distributions is the population balance model (PBM). The population balance concept is very similar to other conservation laws, e.g., mass, heat, and momentum. The main difficulty with the population balance technique is in finding acceptable models for describing the birth and death of the bubbles [1, 2], i.e., coalescence and breakup kernel functions are essential to solve a population balance model [3].

The PBM was first formulated for chemical engineering purposes by Hulburt and Katz [4]. Mihail and Straja [5] used the population balance model to analyze the one-dimensional bubble size distribution in a bubble column. Lee et al. [6] applied the population balance method for non-coalescing systems, where the bubble escape frequency, or bubble rising velocity, was assumed to be constant for all bubble sizes. Hesketh et al. [7] studied bubble breakup in pipeline flows at very low gas fractions using the population balance method where it was specifically assumed that the breakage frequency and efficiency were independent of the bubble size. The PBM has drawn unprecedented attention from both academic and industrial quarters in the past few years because of its applicability to a wide variety of particulate processes [2]. Much work has been performed using the PBM to describe and simulate droplets in liquid-liquid systems and bubbles in gas-liquid systems, which have demonstrated the great potential of PBM for such systems [3, 8].

Prediction of the bubble size and its distribution in oil-based systems is essential and useful in bio-desulphurization processes in the oil industry. In the present study, the local bubble size distribution was predicted using a one-dimensional PBM. Proper coalescence and breakup rates were used for water- and oil-based systems. The model was improved with a fixing parameter in order to reduce the deviation between model prediction and experimental data.

2 Model Description

The PBM is a statistical model that describes the size distribution of the dispersed phase in a multiphase system. The general equation for the PBM applied to the bubbles in a bubble column can be expressed as [1][1)]:

$$\frac{\partial n_i}{\partial t} + \nabla.\overline{u_i} n_i = BC + BB - DC - DB \quad (1)$$

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1) List of symbols at the end of the paper.

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where BC, BB, DC and DB are the rates of birth and death of bubbles due to coalescence and breakage per unit volume for bubbles of size d_i at time t, respectively.

In the present study, only the change of bubble number density in the axial direction at the steady-state condition was considered. In order to keep the model simple and convenient, the dispersion in a bubble column can be considered as a series of well-mixed cells such that the bubble number density can be considered to be uniform in each cell. Therefore, BC, BB, DC and DB for the bubbles of size v_i for each cell would only be functions of v_i and n_i. Replacing these rates in Eq. (1) by the expressions developed by Luo [1] results in the expression given in Eq. (2):

$$\frac{(u_g/\varepsilon_g)_{in}}{\Delta H}(n_i - n_{i,in}) = \sum_{v_j=v_{min}}^{v_i/2} \Omega_c(v_j : v_i - v_j) - \sum_{v_j=v_{min}}^{v_{max}-v_i} \Omega_C(v_j : v_i) + \sum_{v_j=v_i}^{v_{max}} \Omega_B(v_j : v_i) - \Omega_B(v_i) \quad (2)$$

In order to solve Eq. (2), the bubble coalescence rate, Ω_c, the bubble breakup rates, $\Omega_B(v_i)$ and $\Omega_B(v_j{:}v_i)$ should be determined. The calculation methods for these parameters are discussed in the following sections.

Eq. (1) should be solved subject to proper boundary conditions. In the present work, a constant initial bubble diameter at the sparger was considered as the boundary condition. This value was determined by the experimental results at each operating condition. Moreover, the gas holdup in the model was considered to be constant in each operating condition, which was also determined by the experiments performed in this work.

2.1 Bubble Coalescence

Collision or coalescence in bubble columns occurs due to different mechanisms such as turbulence, shear stress and buoyancy [1, 9, 10]. The turbulence mechanism is due to turbulent collisions of bubbles in their random motion, the shear stress mechanism is due to viscous or shear collisions and the buoyancy mechanism is due to the difference in the rise velocities of the bubbles of different sizes [10, 11]. In the present work, only coalescence due to buoyancy was considered, since its effect is more profound compared to the other mechanisms. The bubble coalescence rate due to buoyancy is given as [10]:

$$\Omega_c(v_i, v_j) = \frac{\pi}{4}(d_i + d_j)^2 |u_{ri} - u_{rj}| P_c(v_i, v_j) \quad (3)$$

where the rise velocity of bubbles, $u_{ri,j}$, can be calculated using the Prince and Blanch equation [10] and the coalescence efficiency, P_c, can be calculated using the model of Luo [1].

2.2 Bubble Breakup

In the present work, the following assumptions were considered for modeling the bubble breakage rate:

I. In the fully developed turbulent flow, bubble breakup is mainly caused by eddy collision [1, 3].
II. Only binary breakup is considered, since binary breakup is the main mode of breakup supported by experimental observations of Hesketh et al. [12].
III. Bubble breakup is caused by collision with turbulent eddies of a size equal to or smaller than the bubble size and having sufficient kinetic energy. Eddies larger than bubbles merely transport the bubbles [8].
IV. A local minimum probability exists at exactly equal breakup since more energy is required for binary equal-sized breakup than binary unequal-sized breakup. Therefore, the relative probability of creating daughter bubbles of exactly the same size should not be zero. The above observations are also supported by the experimental results of Hesketh et al. [12].
V. The daughter bubble size distribution depends on the mother bubble size and the energy dissipation rate [3].

Based on the above assumptions, the breakup kernel function can be calculated by [1, 3]:

$$\Omega_B(v_i : v_i f_{Bv}) = 0.923(1-\varepsilon_G) n_i \left(\frac{\varepsilon}{d_i^2}\right)^{\frac{1}{3}} \int_{\xi_{min}}^{1} \frac{(1+\xi)^2 e^{-\chi_c}}{\xi^{\frac{11}{3}}} d\xi \quad (4)$$

where

$$\xi_{min} = \frac{\lambda_{min}}{d_i} \quad (5)$$

$$\xi = \frac{\lambda}{d_i} \quad (6)$$

$$\chi_c = \frac{12 c_f}{We_i} \xi^{-11/3} \quad (7)$$

$$c_f = f_{Bv}^{2/3} + (1 - f_{Bv})^{2/3} - 1 \quad (8)$$

$$We_i = \frac{\rho_L d_i \bar{u}_i^2}{\sigma} \quad (9)$$

$$\bar{u}_i = \sqrt{2.41}(\varepsilon d_i)^{1/3} \quad (10)$$

and

$$\Omega_B(v_i) = \frac{1}{2}\int_0^1 \Omega_B(v_i : v_i f_{Bv}) d f_{Bv} \quad (11)$$

where λ_{min} is the minimum eddy size for the bubble breakup.

The models and experimental data on breakup indicate that the energy dissipation rate, e, has significant influence on the bubble size distribution. The energy dissipated by a bubble is calculated as the product of the drag force and the slip velocity since the drag force is balanced with the buoyancy. The energy dissipation rate can then be calculated as follows [1]:

$$\varepsilon = \frac{(\rho_L - \rho_g) g \sum N_i v_i u_{slip,i}}{\rho_L(1-\varepsilon_g)} = \frac{\Delta\rho g u_g}{\rho_L(1-\varepsilon_g)} \quad (12)$$

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3 Experimental

3.1 Apparatus and Materials

A schematic diagram of the experimental setup used in this work is shown in Fig. 1. The bubble column was made of a pyrex column of 1.1 m height and 0.095 m inside diameter. The experiments were carried out in a semi-batch operation mode where the air continuously flows through the sparger into the fluid-filled column. The air flow rate was measured by a rotameter to provide a gas velocity in the range of 0.098–0.65 cm/s. The sparger was a 0.042 m diameter perforated plate with twenty six 0.3 mm holes and a pinch of 0.5 cm. The sparger was located 0.07 m above the base of the column.

The liquids employed in the present work are listed in Tab. 1. Mixture A was prepared by mixing kerosene and Isomax-Diesel (Tehran Oil Refinery Co., Iran) and mixture B by mixing kerosene and lube oil HVI-150 (Behran Co., Iran). The physical properties of the oil-based liquids were measured at 25 °C. The surface tension was measured by the ring method using a tensiometer (Model K10ST, KRUSS GmbH, Hamburg, Germany), and the viscosities were measured using a viscometer (MCR300-Physica, Germany).

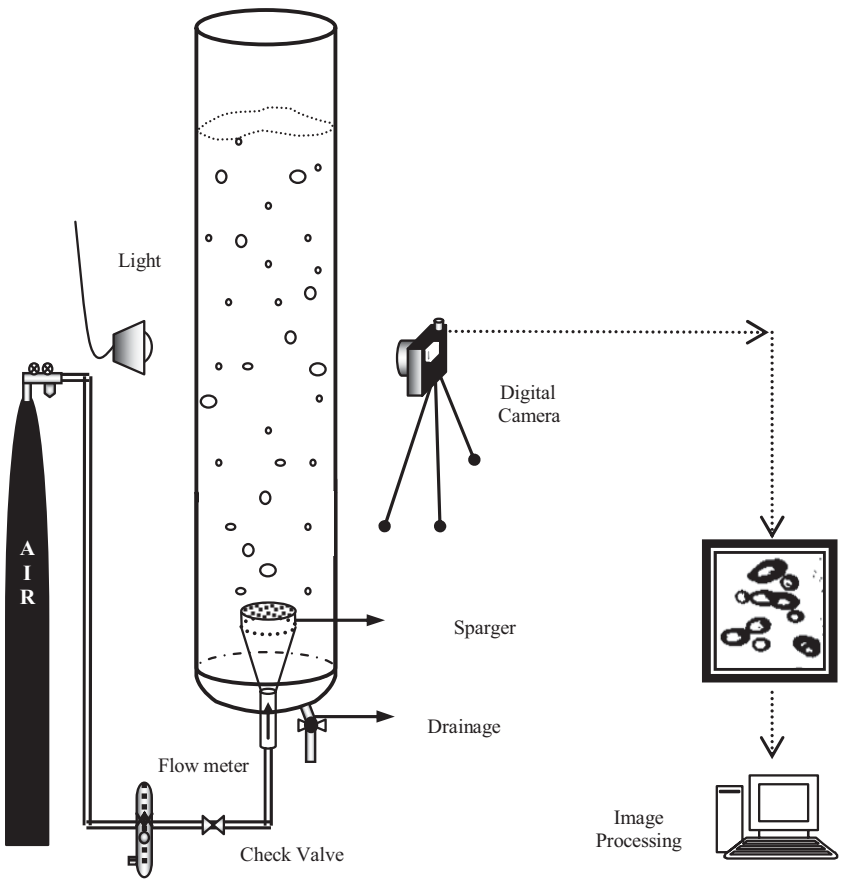

Figure 1. Experimental setup.

Table 1. Physical properties of the liquids used in this work.

Liquid	Density (kg/m^3)	Kinematic Viscosity (10^{-6} m^2/s)	Surface Tension (mN/m)
Water	997	1.003	69.2
Kerosene	782.66	2.17	25.8
Mixture A	799.89	4.76	24.8
Mixture B	799.85	4.45	26.4

3.2 Measurements

The measurement of the bubble sizes and their distributions were performed in the bubble column using a photographic technique. The sizes of not less than two hundred bubbles were measured in each experiment. At the steady-state condition, images of bubbles were photographed using a digital CCD camera (Canon PowerShot-SI3) at three different heights, i.e., 0.22 m (bottom section), 0.52 m (middle section) and 0.82 m (top section) from the base of the column. The measuring method was performed semi-automatically, using a computer program developed during this work. The correction to the real size was based on the scale attached to the column, which was at the same focal distance as the measured bubbles. In fact, the focus was adjusted on the scale and only the well-focused bubbles were measured. In case of ellipsoidal bubbles, the major and minor axes of the bubble images were measured. The equivalent bubble size, equal to the diameter of a sphere whose volume is equal to that of the bubble, was then calculated [13].

The overall gas holdup was determined by the volume expansion method [14]. The unaerated and aerated liquid heights were measured and holdup was then calculated using Eq. (13):

$$\varepsilon_g = \frac{H_D - H_L}{H_D} \tag{13}$$

4 Results and Discussion

In order to validate the population balance model, the calculated bubble size distributions for the air-water system and petroleum-based systems, i.e., air-kerosene and oil-based mixtures at different gas velocities, were compared with the correspond-

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Table 2. Initial bubble diameters in mm.

System	Velocity (cm/s)			
	0.1	0.235	0.352	0.59
Air-water	3.5	–	–	5.5
Air-kerosene	3.5	4.5	5	5.5
Air-A mixture	4	–	4.5	5
Air-B mixture	4	–	5	5.5

ing experimental data. The experimental boundary conditions and gas holdups used in this work for solving the model equations are listed in Tab. 2. The experimental bubble size distributions of the air-water system at different superficial gas velocities are shown in Figs. 2a)–2d). The corresponding prediction of these distributions by the model of Lou [1] is also shown in Fig. 2. It is obvious from Figs. 2a)–2d) that the model of Luo [1], as described in the previous section, is not able to predict the experimental data satisfactorily. In fact, the bubbles in this system are smaller than those predicted by the model of Luo [1]. This difference can be lowered by increasing

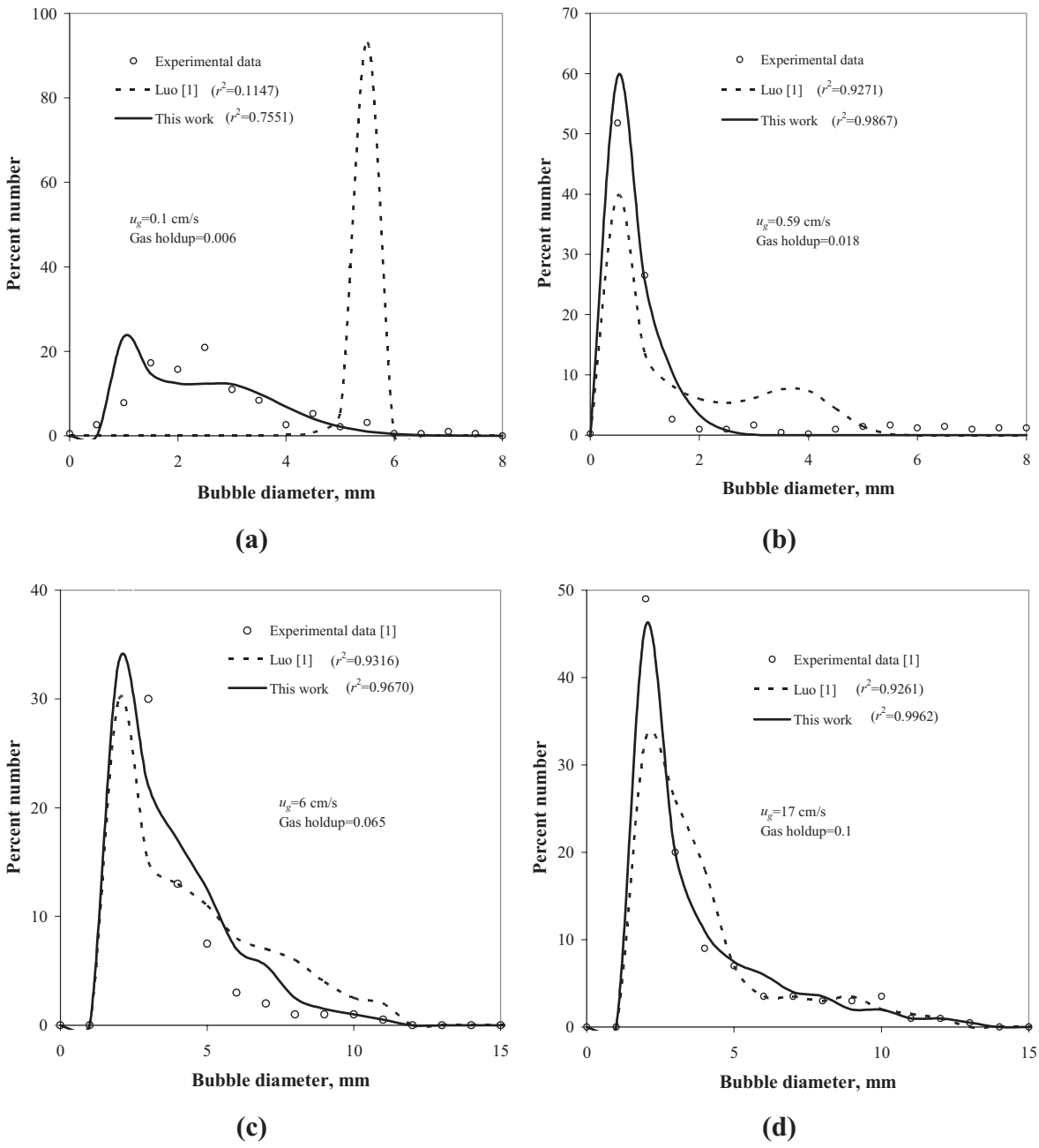

Figure 2. Bubble size distribution in the air-water system: (a) $u_g = 0.1$ cm/s, (b) $u_g = 0.59$ cm/s, (c) $u_g = 6$ cm/s, and (d) $u_g = 17$ cm/s.

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the gas velocity, which increases the turbulence of the system, and thus, increases the energy dissipation rate. Therefore, it can be suggested that the actual energy dissipation of this system is higher than that predicted by Eq. (12).

Based on the above discussion, in order to overcome the deficiency of the model and improve the model predictions, it was suggested in the present work that the energy dissipation rate should be corrected by applying a correcting factor to Eq. (12), as follows:

$$\varepsilon = a\frac{\Delta\rho g u_g}{\rho_L(1-\varepsilon_g)} \tag{14}$$

in which a is the correction factor and could be evaluated by fitting the experimental data to the model. This factor was found by applying the least squares technique to this problem to minimize the sum of errors between the model and experimental data. The results of the corrected model for bubble size distribution (using the correction factor) are also shown in Figs. 2a)–2d). It can be seen in these figures that applying the correction factor has considerably improved the model prediction, especially at lower gas velocities where the actual energy dissipation rate is lower than that predicted by Eq. (12).

Figs. 3a)–3d) show the experimental bubble size distribution of the air-kerosene system at four different superficial gas

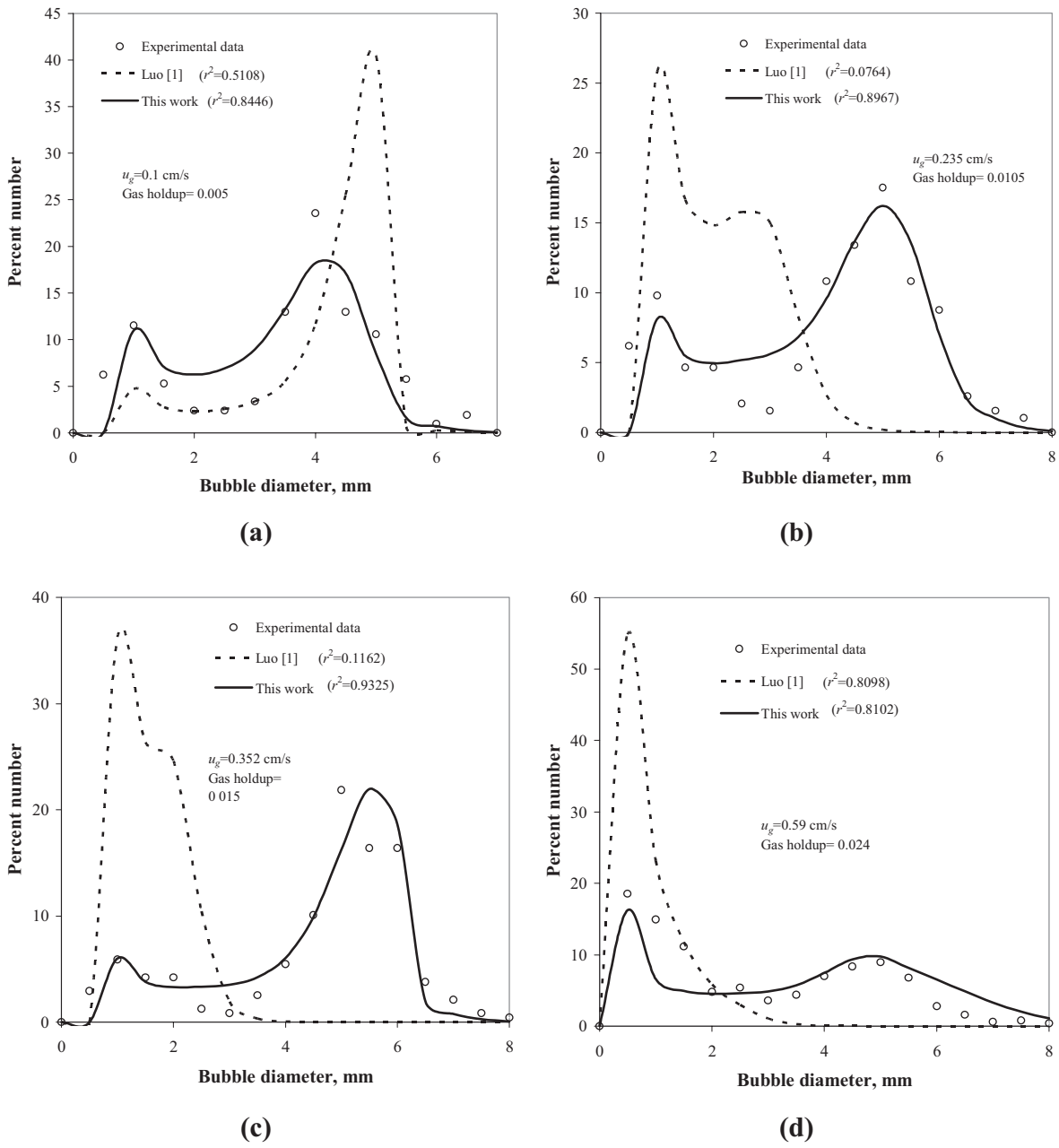

Figure 3. Bubble size distribution in the air-kerosene system: (a) u_g = 0.1 cm/s, (b) u_g = 0.235 cm/s, (c) u_g = 0.352 cm/s, and (d) u_g = 0.59 cm/s.

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velocities. The predictions of the Luo's model [1] and the corrected model for this work are also illustrated in these figures. It can be observed in Figs. 3a)–3d) that increasing the superficial gas velocity in this system results in increasing the error of the Luo's model [1]. In fact, Luo's model [1] underpredicts the size of the bubbles and this deviation increases with increasing gas velocity. In other words, the bubble breakage rate is lower than that predicted by the model of Luo [1]. This trend is the opposite of what was observed for the air-water system in Figs. 2a)–2d) Such a difference in trend could be contributed to the difference in viscosities of the two liquids. The kinematic viscosity of kerosene is higher than the kinematic viscosity of water. Hence, the breakage rate in the air-kerosene system would be lower than that in the air-water system. In a similar manner to what was suggested for the air-water system, correcting the energy dissipation rate using the correction factor introduced in Eq. (14), would improve the prediction of the model. The results of the corrected model are also shown in Figs. 3a)–3d) and confirm that this correction has improved the performance of the model.

The bubble size distributions of the air-mixture A and air-mixture B systems at different superficial gas velocities, are shown in Figs. 4a)–4c) and Figs. 5a)–5c), respectively. The results of the models of Luo [1] and this work are also shown in

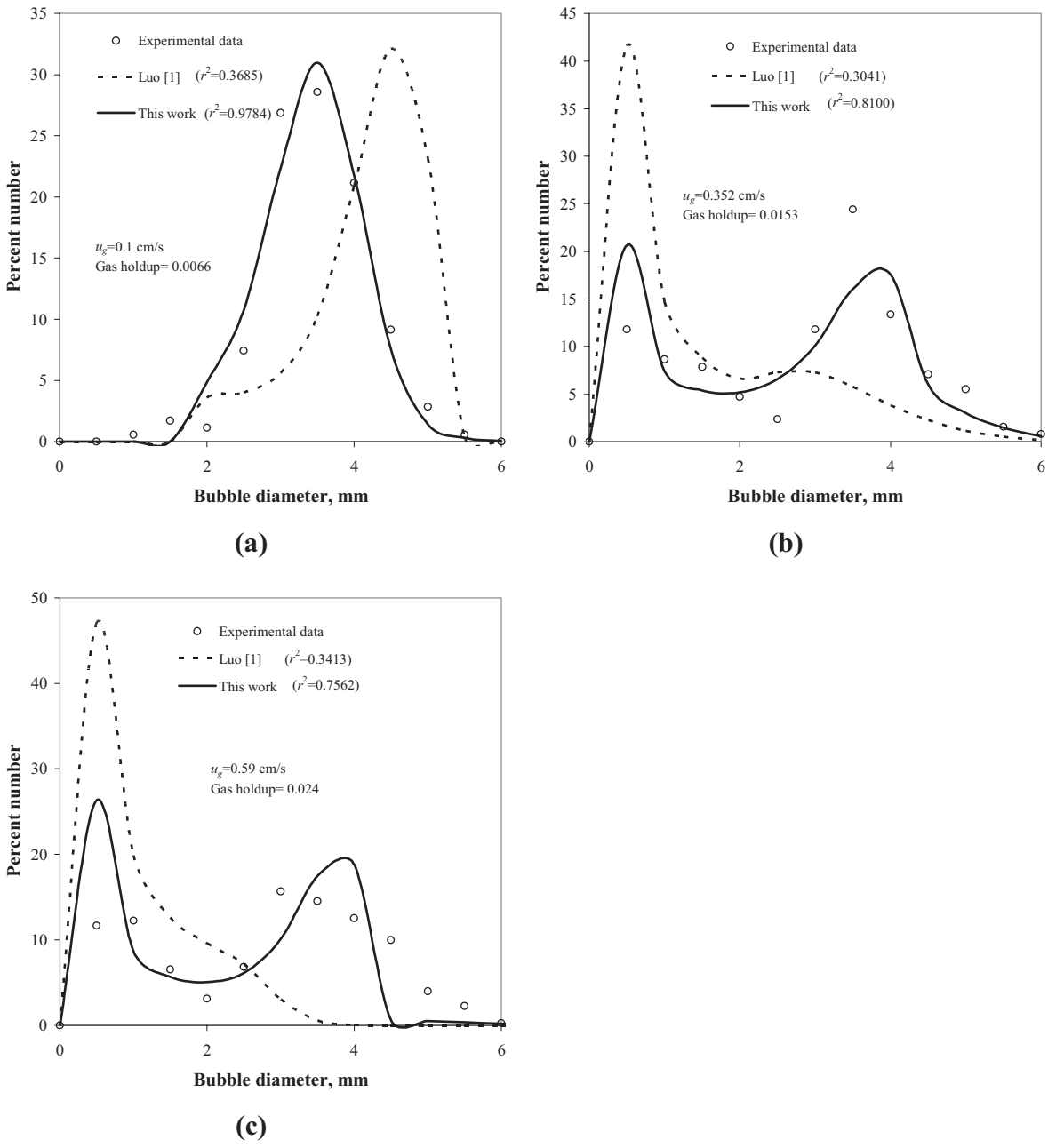

Figure 4. Bubble size distribution in the air-mixture A system: (a) $u_g = 0.1$ cm/s, (b) $u_g = 0.352$ cm/s, and (c) $u_g = 0.59$ cm/s.

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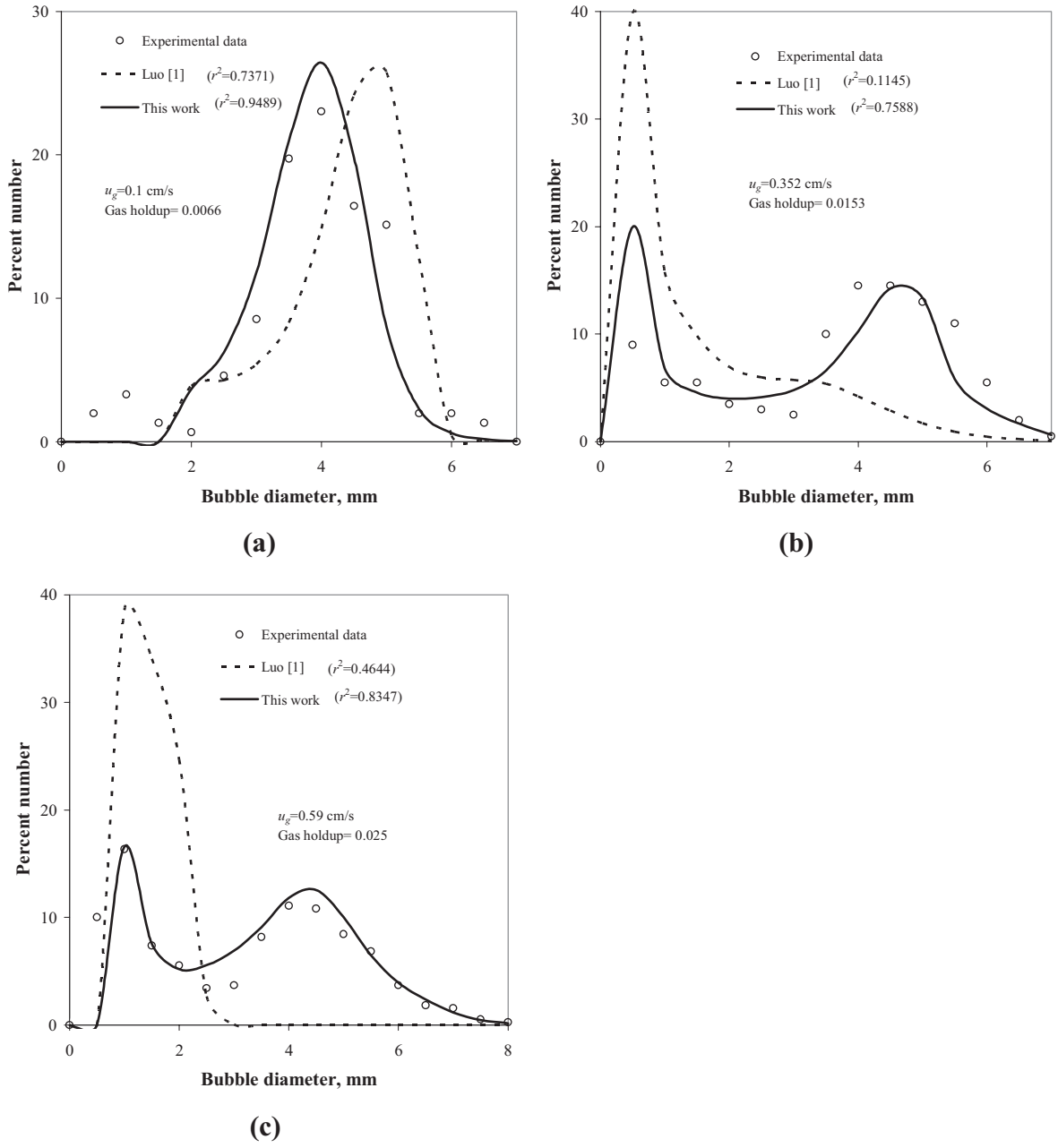

Figure 5. Bubble size distribution in the air-mixture B system: (a) $u_g = 0.1$ cm/s, (b) $u_g = 0.352$ cm/s, and (c) $u_g = 0.59$ cm/s.

these figures. These figures also indicate that increasing the gas velocity results in the deviation of the model of Luo [1] from the experimental data increasing, which indicates that this model fails to correctly predict the breakage rate. As discussed in Fig. 3, at higher viscosities, the breakage rate is lower than what would be predicted by the model of Luo [1] and correcting the energy dissipation rate by the factor introduced in Eq. (14) has considerably improved the results of the model.

The calculated values of the correction factor, *a*, which were achieved by the least squares method, are illustrated in Fig. 6. It can be seen in Fig. 6 that the correction factor decreases on increasing the superficial gas velocity in all systems. This trend suggests that the energy dissipation rate is higher than that given by Eq. (12) at low gas velocities while it is lower than that given at higher gas velocities. This correction factor is higher for low viscosity liquids.

5 Conclusions

A population balance model from the literature was adopted for estimating the bubble size distribution in bubble columns. This model was applied to the air-water system as well as three other systems containing petroleum-based liquids. For systems

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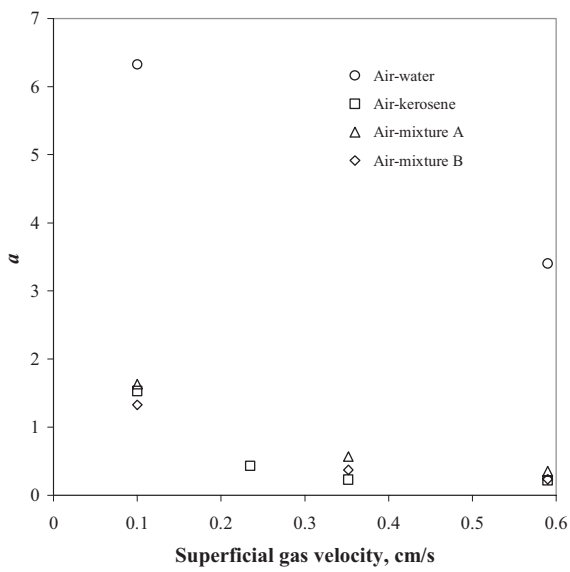

Figure 6. Correction factor for different systems.

with low kinematic viscosity, e.g., the air-water system, increasing the superficial gas velocity in the systems will increase the breakage process and the number of small bubbles. The resultant error between the model prediction and the experimental data for these systems will decrease with increasing superficial gas velocity. Due to the higher kinematic viscosity of the oil-based systems, increasing the superficial gas velocity leads to increases in the resultant errors. The energy dissipation rate is one of the parameters which affect the bubble size distribution and an investigation of this parameter was carried out. Thus, a fixing parameter was introduced in the energy dissipation rate equation to reduce the resultant errors between the modeling prediction and experimental data. Good agreement was achieved between the model prediction and experimental results when the fixing parameter was applied.

Symbols used

a	[–]	correction factor
BC	[1/(m^{3}s^{-1})]	birth rate of bubbles because of coalescence process
BB	[1/(m^{3}s^{-1})]	death rate of bubbles because of coalescence process
c_f	[–]	parameter of increase of surface energy
DC	[1/m^{3}s^{-1}]	birth rate of bubbles because of breakage process
DB	[1/m^{3}s^{-1}]	death rate of bubbles because of breakage process
d	[m]	bubble diameter
f_{BV}	[–]	volume breakage fraction
g	[m/s^2]	gravitational acceleration
H	[m]	dispersion height
n	[1/m^3]	number density of bubbles
N	[1/m^3]	bubble number per volume
P_c	[–]	coalescence efficiency
r^2	[–]	Pearson correlation coefficient
t	[s]	time
$\bar{u}_G$	[m/s]	superficial gas velocity
u_i	[m/s]	turbulent velocity
u_r	[m/s]	bubble rise velocity
u_{slip}	[m/s]	bubble slip velocity
v	[m^3]	bubble volume
v_{min}	[m^3]	minimum bubble volume
v_{max}	[m^3]	maximum bubble volume
We	[–]	Weber number

Greek symbols

e_g	[–]	gas hold up
e	[m^2/s^3]	energy dissipation rate
λ	[m]	eddy size
λ_{min}	[m]	minimum eddy size effective for breakage
ρ	[kg/m^3]	density
σ	[N/m]	surface tension
Ω_B (v_i)	[1/s]	bubble breakage rate of a bubble with size v_i
Ω_B ($v_i v_i f_{Bv}$)	[1/s]	breakage rate of a bubble with size v_i breaking with breakage fraction f_{Bv}
Ω_C ($v_i v_j$)	[1/s]	coalescence rate of a bubbles with size v_i and v_j

Subscripts

D	dispersed phase
g	gas phase
i,j	referring bubbles
in	input
L	liquid phase

References

[1] H. Luo, *Ph.D. Thesis*, University of Trondheim, Norway **1993**.
[2] D. Ramkrishna, A. W. Mahonay, *Chem. Eng. Sci.* **2002**, *57*, 595.
[3] T. Wang, J. Wang, Y. Jin, *Chem. Eng. Sci.* **2003**, *58*, 4629.
[4] H. Hulburt, S. Katz, *Chem. Eng. Sci.* **1964**, *19*, 555.
[5] R. Mihail, S. Staraj, *Chem. Eng. J.* **1986**, *33*, 71.
[6] C. H. Lee, L. E. Erickson, L. A. Glasgow, *Chem. Eng. Commun.* **1987**, *59*, 65.
[7] R. P. Hesketh, A. W. Etchells, *Chem. Eng. Sci.* **1991**, *46*, 1.
[8] F. Lehr, D. Mewes, *Chem. Eng. Sci.* **2001**, *56*, 1159.
[9] X. Y. Fu, M. Ishii, *Nucl. Eng. Des.* **2002**, *219*, 143.
[10] M. J. Prince, H. W. Blanch, *AIChE J.* **1990**, *36*, 1485.
[11] S. K. Friedlander, *Smoke, Dust and Haze*: *Fundamentals of Aerosol Behavior*, John Wiley and Sons, New York **1977**.
[12] R. P. Hesketh, A. W. Etchells, T. W. Fraser Russell, *Ind. Eng. Chem. Res.* **1991**, *30*, 835.
[13] M. Pollia, M. D. Stanislaoa, R. Bagatinb, E. Abu-Bakra, *Chem. Eng. Sci.* **2002**, *57*, 197.
[14] L. Z. Pino et al., *Chem. Eng. Commun.* **1992**, *117*, 367.

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